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FIRST SEMESTER B.Tech EXAMINATION, JANUARY-2025

RELATIONAL DATABASE MANAGEMENT SYSTEMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		Questions	Blooms Level	CO's	Marks
		PART-A	**5 x2=10**		
1.	**a.**	What is three-schema architecture?	L1	CO1	**02**
	b.	**State** the function of aggregate operations in relational algebra	L1	CO1	**02**
	c.	Differentiate **WHERE and HAVING** clauses in SQL.	L2	CO1	**02**
	d.	List two advantages of using **schema change statements**.	L1	CO1	**02**
	e.	Define **Functional Dependency** in the context of database normalization.	L1	CO1	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain database languages like DDL (Data Definition Language) and DML (Data Manipulation Language), and provide real-world use cases for both.	L3	CO2	**05**
		OR			
	b.	Describe how the database approach addresses the limitations of the traditional file-based approach.	L2	CO2	**05**
3.	**a.**	**Analyze** the difference between unary and binary operations in relational algebra using suitable examples.	L4	CO3	**05**
		OR			
	b.	**Describe** the process of converting an ER diagram into a relational schema with an example.	L2	CO3	**05**
4.	**a.**	Differentiate **GROUP BY and ORDER BY** clauses in SQL. Provide examples showing how they affect query results.	L2	CO3	**05**
		OR			
	b.	Explain the concept of constraints in SQL. Discuss how to apply **NOT NULL, UNIQUE, and CHECK** constraints in a table definition.	L2	CO3	**05**
5.	**a.**	Write a SQL query to create a **VIEW** that combines data from two tables: **Employees and Departments**.	L2	CO4	**05**
		OR			
	b.	Illustrate the syntax of an **action trigger** to log changes in a table	L3	CO4	**05**

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		called **Orders.**			
6.	**a.**	Explain the difference between Boyce-Codd Normal Form (BCNF) and Third Normal Form (3NF) with an example.	L2	CO5	**05**
		OR			
	b.	Write SQL query to demonstrate **transaction support**, including BEGIN TRANSACTION, COMMIT, and ROLLBACK statements.	L3	CO6	**05**
7.	**a.**	Write SQL statements to create a database and a table named **Employees** with columns for **EmployeeID, Name, Department, Salary**, and **JoinDate.** Specify appropriate data types and constraints, including primary key and NOT NULL constraints.	L3	CO4	**05**
		OR			
	b.	Create a table **Orders** with columns **OrderID, CustomerID, OrderDate**, and **TotalAmount,** write SQL queries to: (i) retrieve all orders placed in the last month, (ii) sort the results by **TotalAmount** in descending order.	L3	CO4	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Create a conceptual design for a hospital database system. Include entity types, relationships, and attributes, focusing on patient records, doctors, appointments, and treatments. Provide an ER diagram.	L4	CO1	**10**
		OR			
	b.	Using SQL, implement a database schema based on the ER diagram of a bookstore. Include tables, attributes, primary keys, and foreign keys, along with sample SQL queries for retrieving data.	L3	CO2	**10**
9.	**a.**	Design a relational schema for an online retail database, including tables, keys, and constraints. Create sample relational algebra queries to retrieve customer orders, total sales per product, and customers who have made purchases above a certain amount.	L4	CO3	**10**
		OR			
	b.	Evaluate the process of mapping a conceptual ER design to a logical relational schema. What are the potential pitfalls, and how can they be mitigated? Provide examples.	L4	CO3	**10**
10.	**a.**	Analyze the use of constraints in SQL table definitions, such as **PRIMARY KEY, FOREIGN KEY, CHECK**, and **UNIQUE.** How do these constraints ensure data integrity? Provide examples.	L4	CO3	**10**
		OR			
	b.	Critically assess the role of Data Definition Language (DDL) in database schema evolution. How does it help in managing database changes,	L3	CO3	**10**

		and what are the challenges faced during schema alterations? Provide examples.			
11.	**a.**	Write a stored function in SQL to calculate the yearly bonus (10% of salary) for each employee and call this function within a query.	L3	CO4	**10**
		OR			
	b.	Create a stored procedure that accepts two inputs (Employee ID, New Salary) and updates the salary only if it's within a permissible range. Explain the flow of logic.	L4	CO4	**10**
12.	**a.**	Compare the different **normal forms** (2NF, 3NF, BCNF, 4NF, and 5NF) based on their requirements and constraints. Provide examples for each.	L4	CO5	**10**
		OR			
	b.	Write a detailed SQL example demonstrating a **transaction** that updates multiple tables, ensures serializability, and recovers correctly after a failure.	L3	CO6	**10**
13.	**a.**	Evaluate the effectiveness of using SQL's **INSERT, UPDATE**, and **DELETE** commands for maintaining data consistency in a multi-user environment. Discuss how transaction control statements like **COMMIT** and **ROLLBACK** enhance data reliability.	L4	CO4	**10**
		OR			
	b.	Propose a set of SQL queries for a sales database that retrieves insights such as total sales per region, top-selling products, and customer purchase frequency. Create the necessary tables, insert sample data, and provide the queries with results.	L4	CO4	**10**